Juan Ignacio Ogdon

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Clickable links: LinkedIn | GitHub Portfolio

Profile

Aerospace Engineering student (6th year, expected graduation Dec 2026) at UTN, focused on control systems, genetic algorithms, and neural network-based controllers for autonomous aerospace applications. Experienced in spacecraft mechanisms, rocket simulation, and international research collaboration. Seeking graduate research opportunities in advanced control and AI-based systems.

Education

- **Universidad Tecnológica Nacional (UTN)** — Expected Dec 2026
  B.Sc. Aerospace Engineering (6-year degree) | GPA: 8.1/10 — *Buenos Aires, Argentina*
- **Don Bosco High School** — Dec 2020
  Diploma in Economics and Administration | GPA: 9.24/10 — *San Carlos de Bariloche, Argentina*

Experience

- **Universidad Tecnológica Nacional (UTN)** — 2025 – Present
  Teaching Assistant — Control Systems (click to see more) — *Buenos Aires, Argentina*
  - Focused on control systems for aerospace applications, with emphasis on AI-based control methods, simulation, and system dynamics.
- **Private Tutor — Mathematics & Physics (click to see more)** — 2019 – Present
  Independent — *Buenos Aires, Argentina*
  - One-on-one tutoring over 7+ years; sustained student referrals reflect consistent teaching effectiveness.
  - Supported students with diverse learning needs, helping them achieve significant academic improvements.
- **LIA Aerospace** — 2024 – 2025
  Intern — *Buenos Aires, Argentina*
  - Worked designing ground support equipment for liquid rocket engine.
- **Grupo de Tecnología Aeroespacial, UTN** — 2022 – 2024
  Research Student — *Buenos Aires, Argentina*
  - Developed a CubeSat appendage release system for solar panels; awarded **Best Presentation in Satellite Technology** at the IAA Latin American Small Satellite Conference (2024).

Interests

- Aerospace applications of AI
- Rocket propulsion and control systems
- Autonomous systems and drones
- FEA and Structural analysis
- Space systems engineering
- Computational Fluid Dynamics (CFD)

Selected Projects (click to see more)

- **GA-Evolved Neural Network Rocket Landing Controller** — Trained a minimal neural network to solve 2D TVC rocket landing using a genetic algorithm pipeline. Features adaptive curriculum staging, island-model population diversity, and physics-derived anti-exploit fitness design across three rocket families.
- **CubeSat Appendage Release System** — Design, simulation, and characterization of a solar panel deployment mechanism; awarded Best Presentation at IAA 2024.
- **Rocket Attitude Control Simulation** — Simulated thrust vector control rocket dynamics using Simulink.

Skills

- **Programming & Simulation:** MATLAB, Simulink, Mathematica, Python, OpenFOAM
- **Engineering Tools:** CAD Design, FDM 3D Printing, Optimized Design for 3D Printing, FEA
- **Collaboration:** Atlassian Jira, Confluence; team-based research projects
- **Soft Skills:** Strong communicator and collaborative problem-solver; adaptable across research, engineering, and teaching environments; quick, self-directed learner.

Languages

- Spanish: Native | English: Bilingual (C1 certified)